

Maria Esther Hernandez Ortega

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Portafolio en GitHub: <https://github.com/38195214-cell> CV digital: <https://38195214-cell.github.io/>

Recently graduated Robotics and Mechatronics Engineer with strong knowledge in embedded systems development, microcontroller-based device control, and process automation. Recognized for responsibility, teamwork, effective communication, and commitment to continuous improvement. Passionate about technological innovation and focused on applying engineering knowledge to develop efficient and high-impact solutions.

PROFESSIONAL EXPERIENCE

Labsol(Free Software Laboratory) Robotics Project Development – Social Service	Zacatecas, Zacatecas October 2024–May 2025
- Developed a robotics project for the Free Software Laboratory. - Implemented an interactive RGB LED racing system using stationary bicycles through encoder signal processing. - Integrated sensors, microcontrollers, and real-time control programming.	
VEIMAZ (Agricultural Machinery Workshop) Full Stack Developer – Engineering and Automation	Col. Hidalgo, Sombrerete, Zacatecas May 2025 – August 2025
- Operated and controlled CNC plasma cutting machines. - Designed mechanical parts and cutting paths using AutoCAD. - Participated in automation and manufacturing processes.	
Johnson Electric Engineering Intern – Innovation Engineering	Calera, Zacatecas January 2026 – June 2026
- Developed a dynamometer project integrating mechanical, electronic, and software systems. - Designed mechanical components using SolidWorks. - Developed and tested code for project systems. - Participated in multidisciplinary hardware and software integration. - Completed technical training through specialized Udemy courses focused on software development and engineering.	

EDUCATION

AUTONOMOUS UNIVERSITY OF ZACATECAS Bachelor’s Degree in Robotics and Mechatronics Engineering	Zacatecas, Zacatecas 2021-2026 (Degree in process)
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CERTIFICATIONS

Association of Applied Robotics and Technology Sciences
Advanced Course in Applied Robotics – STPS Certified
Excellence Diploma - Python Course – STPS Certified

TECHNICAL SKILLS

• Python	• AutoCAD	• C, C++	• Microsoft Office	• Edge Impulse
• MicroPython	• SolidWorks	• Sensores	• Excel, Word, etc.	• Microcontroladores
• Verilog(FPGA)	• Proteus	• Actuadores	• Microsoft Azure	• Git / GitHub
• HTML	• Altium Designer	• Visual Studio	• MATLAB	• Node-RED

LANGUAGES

- Español – Nativo
- Inglés – Intermedio (B2)

FEATURED PROJECTS

Smart Home Fire Detection and Control System

- Developed an IoT-based system for home fire monitoring and control.
- Implemented cloud-connected sensors for continuous real-time data transmission.
- Integrated SQL databases for data storage and processing.
- Programmed a microcontroller for automatic decision-making and system response activation.

Physical Activity Monitoring System with Artificial Intelligence

- Developed and implemented an artificial intelligence model for physical activity recognition.
- Trained a model capable of identifying activities such as walking, running, and climbing stairs.
- Integrated motion sensors on X, Y, and Z axes with a microcontroller.
- Estimated steps and floors climbed through motion data processing.

Fixture Design for Assembly Validation

- Designed and developed a mechanical fixture for assembly distance validation.
- Project developed for the process engineering area at Johnson Electric.
- Created mechanical models and designs using SolidWorks.
- Focused on dimensional validation and industrial process improvement.